

Influence of load and carrying methods on gait phase and ground reactions in children's stair walking

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Abstract

The study investigated the effects of carrying methods and loads on gait phase and ground reaction force during stair ascent and descent in children. The carrying methods that were examined included the backpack and one-strap athletic bag. The load weights included 0%, 10%, 15%, and 20% of body weight. Thirteen school children aged 12.21 ± 0.98 years were recruited as subjects. A Novel Pedar System was used to record and analyze the insole pressure during stair walking with different loads. The load that caused a significant increase in the peak force in each bag and stair mode was 15% of body weight, except for stair ascent carrying the athletic bag, where the load was 10% of body weight. The maximum peak force that was induced by this load in stair descent was 1.25 times that in descent with no load, 1.89 times that in ascent with the same load, and 2.19 times that in ascent with no load. The force-to-time ratio in descent was about three times that in ascent.

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1. Introduction

Both load carriage and stair walking are frequently encountered during the course of normal daily activities. In many schools, children need to carry their schoolbags up and down stairs. The problem of over-weight schoolbags has been of high concern to parents and the community in recent years.

A number of studies of stair walking have been published. The published literatures focused on adults and provided useful information about the kinematics, ground reaction forces, joint kinetics, and muscle dynamics in stair walking [1–6]. To the author's knowledge, however, information about the kinetic characteristics of children during stair walking while carrying loads has not been reported. Hence, the purpose of this study was to investigate the effects of load and carrying methods on ground reaction force and gait temporal characteristics during both stair modes in children.

2. Method

Thirteen male students aged 12.21 ± 0.98 years (body weight, 47.12 ± 9.69 kg; body height, 159.66 ± 9.67 cm) were recruited from a secondary school to serve as subjects. All of the subjects were free of injury during the test and did not have injury histories that had caused abnormal gaits or an inability to climb stairs. A consent form was signed and body weight and body height were recorded before the test.

The carrying methods compared in this study were carrying a one-strap athletic bag (across the right shoulder, the dominant side of all subjects) and a backpack (on both shoulders), which represented asymmetrical and symmetrical load carrying methods, respectively. Four loads, at 0%, 10%, 15%, and 20% of the subjects' body weight, were used. The percentage of body weight, rather than absolute weight, was used to classify the carrying load so that normalization across subjects could be achieved. All subjects performed two modes of stair walking. For each mode, a factorial design (four loads $\times$ two bags), i.e., altogether eight separated combinations (trials), was used. The order of the trials was randomized using a Latin square design. The test was

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undertaken in the audience seating of the university's gymnasium. The step dimensions were 16 cm (riser) by 35 cm (tread) with the slope being 25.2° , which is within the recommended stair slope and dimensions of stair riser and tread [7].

An in-shoe pressure measurement system (Novel Pedar System, Germany) was employed to record the temporal and kinetic data during stair walking. Each Novel Pedar insole contained 99 force sensors. All sensors of the insole were individually calibrated with the aid of the trublu calibration device (Novel Pedar System, Germany) before and after testing each day. The subjects were fitted with the proper size insole. The insoles were placed inside both of each subject's shoes to record the vertical reaction forces of each sensor during contact against time. The data collection box was attached to a waistband during the stair walking. The force signals of sensors were recorded at 50 Hz and saved in a PCMCIA card that was installed in the collection box of the system for each trial. Each subject used the same brand of shoes that are most widely worn by the local school students; as the statures of the subjects were similar, only two sizes were worn.

Before each trial, the subjects rested for 5 min. They were then asked to walk on level ground with a load of 20% of body weight for 400 m in free speed to simulate a normal walk to school from home. After the floor walking, the stair walking trial began. In each trial, the subjects climbed the 33-step-staircase, then turned around and walked down to the starting point by following the same path. This was performed three times and three complete gait cycles from the 15th to the 21st step in each ascent and from the 21st to the 15th step in each descent were selected for analysis, leading to nine complete gait cycles from each mode. To reduce the risk of accumulative fatigue affecting performance, the subjects were asked to take a rest after each trial.

The force–time characteristics of walking gait were obtained from the continuous recordings of the pressure insoles. The dependent variables included the peak vertical ground reaction forces, force-to-time ratio of the first peak force, the gait cycle duration, stance duration, single and double support duration, and time to peak force. Only the first peak force was calculated for the force-to-time ratio because as discussed in a previous study [2,5], only this peak

force showed an impulsive nature due to its short development time. All forces were normalized to the individual subject's body weight, while all temporal data were normalized to the gait cycle duration. In each trial, the dependent variables were averaged across the nine complete gait cycles to represent the biomechanical features for each bag and load combination.

Firstly, the paired samples *t*-test was used to investigate whether there was a significant difference in each dependent variable between the left and right foot in each stair mode of each trial. If there was no significant difference, the measures of the left and right feet were pooled and averaged. If, however, there was a significant difference in the variables between the left and right feet, the measures from the foot that showed the greater value were selected for further analysis. All data are presented as mean (S.D.). A repeated ANOVA measurement (four loads $\times$ two bags) was performed to compare each dependent data set. A value of $\alpha = 0.05$ was used for all tests as the criterion to determine the presence or absence of significance.

3. Results

Statistical analysis showed that there was no significant difference between the right and left foot in each dependent variable when a backpack was carried. This allowed an averaging of the dependent variables for the two feet. With athletic bag carrying, the peak force of the left foot was found to be greater than that of the right foot. Hence, the dependent variables of the left foot were chosen for analysis. Moreover, there was no significant difference in the measures between the athletic bag and backpack in the “no load” conditions. Therefore, the bag itself had no influence on the load effects.

In this paper, the gait cycle begins with the initial landing of the leading (right) foot, and ends with the next landing of the same foot. The gait cycle was then divided into sub-phases in accordance with the previous studies [2,4] and illustrated in Fig. 1.

The temporal and kinetic parameters were presented in Tables 1 and 2, respectively. In Table 1, except the “double support” duration that was calculated as the sum of the two

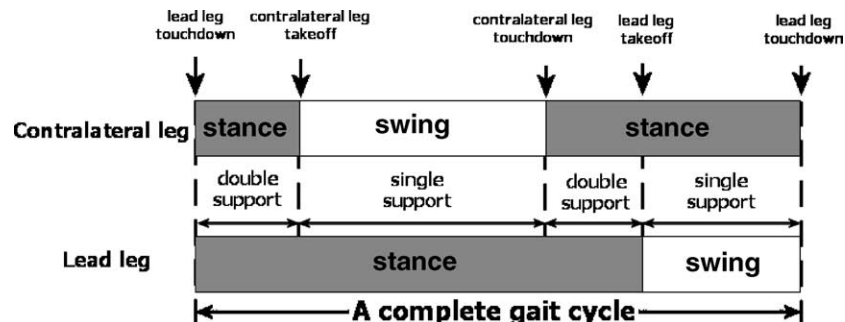


Fig. 1. Sub-phases of one complete gait cycle in stair walking.

Table 1

Mean and S.D. of temporal parameters (gait cycle duration in seconds; other variables in percent of gait cycle duration; $N = 13$)

	Backpack load (%)				Athletic bag load (%)			
	0	10	15	20	0	10	15	20
Ascent								
Time to contralateral takeoff	13.18 (2.63)	13.08 (1.69)	14.70 ^a (2.31)	14.81 ^a (2.91)	12.44 (1.28)	13.28 ^a (1.54)	13.61 ^a (0.83)	14.12 ^{ab} (1.61)
Time to contralateral touchdown	49.65 (1.73)	50.37 (1.78)	49.45 (1.70)	49.18 (2.15)	49.60 (1.92)	49.89 (1.59)	50.71 (1.54)	50.31 (2.18)
Time to lead leg takeoff (stance duration)	62.15 (2.19)	63.39 (1.89)	64.15 ^a (2.41)	64.00 ^a (2.68)	62.04 (1.01)	63.18 (1.47)	64.32 ^{ab} (1.28)	64.43 ^{ab} (1.86)
Time to first peak force	16.66 (2.27)	16.61 (1.94)	17.25 (1.72)	17.24 (1.60)	17.21 (1.89)	18.59 ^a (2.36)	19.86 ^a (2.67)	18.90 ^a (2.00)
Time to second peak force	50.76 (1.96)	49.79 (5.78)	53.01 (2.79)	52.11 (1.94)	49.75 (1.92)	48.79 (5.66)	51.42 (2.71)	51.07 (1.90)
Gait cycle duration	1.20 (0.20)	1.21 (0.10)	1.19 (0.07)	1.25 (0.14)	1.15 (0.09)	1.17 (0.09)	1.18 (0.13)	1.22 (0.12)
Double support duration	25.68 (5.26)	26.17 (3.37)	29.40 ^a (4.63)	29.63 ^a (5.82)	24.88 (2.55)	26.57 ^a (3.08)	27.22 ^a (1.66)	28.24 ^{ab} (3.22)
Descent								
Time to contralateral takeoff	9.62 (1.13)	9.53 (2.22)	10.36 ^a (1.24)	10.38 ^a (1.46)	9.38 (1.90)	9.81 (1.86)	9.99 (1.51)	10.37 (2.23)
Time to contralateral touch down	49.85 (1.44)	49.63 (2.63)	49.26 (1.70)	48.47 (1.80)	49.37 (0.48)	49.75 (1.69)	49.67 (1.92)	49.94 (2.63)
Time to lead leg takeoff (stance duration)	59.46 (1.94)	59.15 (1.91)	59.62 (0.77)	58.85 (1.34)	58.75 (1.72)	59.57 (1.60)	59.65 (1.40)	60.32 (2.01)
Time to first peak force	11.00 (1.63)	11.25 (1.41)	11.88 (1.34)	11.61 (2.02)	12.44 (2.72)	12.30 (2.37)	12.85 (4.05)	12.84 (3.62)
Time to second peak force	51.82 (4.03)	51.73 (4.79)	49.92 (3.83)	48.09 ^{ab} (6.20)	51.56 (4.01)	51.47 (4.76)	49.67 (3.81)	47.85 ^{ab} (6.17)
Gait cycle duration	1.15 (0.18)	1.17 (0.17)	1.11 (0.10)	1.20 ^{ac} (0.21)	1.05 (0.12)	1.07 (0.12)	1.10 (0.16)	1.22 ^{abc} (0.22)
Double support duration	19.23 (2.26)	19.05 (4.45)	20.72 ^a (2.48)	20.76 ^a (2.92)	18.76 (3.81)	19.63 (3.72)	19.97 (3.02)	20.75 (4.47)

All times were calculated from the instant the lead leg touches down.

^a $P < 0.05$ vs. 0%.^b $P < 0.05$ vs. 10%.^c $P < 0.05$ vs. 15%.

“single support” duration, all other parameters were directly measured.

3.1. Ascent

Table 1 shows that carrying an athletic bag with load of 10% of body weight or above induced significant increase in the time from the lead leg touchdown to both the takeoff and touchdown of the contralateral leg, and to the occurrence of the first peak force of the lead leg. This load also increased the total double support duration in the gait cycle. However, the increase in the time from the lead leg touchdown to its takeoff was associated with the load of 15% of body weight or above.

With a backpack, the critical load was 15% of body weight or above that induced significant increase in the time from the lead leg touchdown to both the contralateral leg takeoff and touchdown, and to the lead leg takeoff. The total double support duration under this load was also increased.

The second peak force was significantly greater than the first one for both bags. As shown in Table 2, with the backpack, a load of 10% of body weight or above induced significant increase in the first peak force. At the instants the first and second peak force occurred, the contralateral force, averaged over loads, was 1.59% and 11.95% of body weight, respectively. With the athletic bag, these figures were 0.64% and 12.83% of body weight, respectively. In this carrying method, a load of 10% of body weight significantly increased the second peak force. Comparing the two carrying methods, a load of 20% of body weight induced sig-

nificantly higher second peak force with the backpack than with the athletic bag.

3.2. Descent

As presented in Table 1, a load of 20% of body weight induced a significant increase in gait cycle duration and decreased time to the second peak force for both bags. The significant increase in double support duration and time to contralateral takeoff only occurred with the backpack, and was caused by a 15% of body weight load.

In contrast to stair ascent, during descent, the first peak force was significantly greater than the second one. Table 2 shows that a load of 15% of body weight or above induced a significant increase in both peak forces for both bags, with the exception of the backpack in which the load that induced significant increase in the second peak force was 20% of body weight.

4. Discussion

4.1. Ascent

A load of 15% of body weight induced significant increases in stance and double support duration with the backpack (Table 1). This load for the athletic bag was 10% of body weight. Previous studies of level walking indicated that the duration of the stance phase is one of the sensitive variables of gait; the stance duration increased slightly after loading [8–9]. The load carried on the back raised the center

Table 2
Mean and S.D. of kinetic parameters (force, in percent of body weight; force-to-time ratio, in N/s; $N = 13$)

	Backpack load (%)					Athletic bag load (%)				
	0	10	15	20		0	10	15	20	
Ascent										
First peak force of lead leg	89.21 (14.54)	98.22 ^a (11.82)	103.28 ^{ab} (13.95)	107.67 ^{abc} (18.23)		96.72 (14.89)	99.40 (15.86)	104.38 (15.77)	111.09 (15.61)	
Contralateral force 1	1.56 (2.28)	0.60 (0.95)	1.46 (1.53)	2.72 (4.97)		0.72 (0.64)	0.62 (1.08)	0.48 (0.60)	0.75 (0.78)	
Second peak force of lead leg	109.56 (11.40)	116.81 (16.14)	126.98 ^a (16.76)	138.69 ^{abcd} (13.61)		111.10 (14.32)	118.54 ^a (10.16)	120.15 ^a (9.51)	120.38 ^a (7.19)	
Contralateral force 2	8.22 (7.70)	12.94 (8.73)	11.79 (5.29)	14.84 (9.41)		14.78 (8.87)	10.49 (9.53)	11.22 (12.46)	14.84 (12.32)	
Force-to-time ratio	2164.21 (864.17)	2139.11 (767.96)	2349.31 (1250.82)	2312.78 (1457.22)		2217.91 (835.75)	2151.49 (760.25)	2330.07 (1264.66)	2289.07 (1470.10)	
Descent										
First peak force of lead leg	155.74 (28.77)	165.34 (24.62)	195.09 ^{ab} (16.57)	191.83 ^{ab} (25.28)		166.51 (25.94)	181.10 (42.03)	195.43 ^a (41.74)	181.42 ^a (31.25)	
Contralateral force 1	11.87 (7.91)	16.02 (10.69)	16.80 (8.28)	14.61 (7.77)		12.65 (11.15)	11.73 (9.12)	14.24 (13.98)	13.20 (17.57)	
Second peak force of lead leg	89.58 (11.81)	93.39 (12.57)	93.58 (17.31)	95.4 ^a (11.72)		93.20 (14.33)	91.90 (11.19)	96.52 ^b (15.04)	100.38 ^b (12.57)	
Contralateral force 2	8.64 (8.64)	6.93 (7.03)	12.67 (11.46)	9.81 (13.14)		8.23 (14.05)	9.11 (13.21)	11.48 (17.67)	10.11 (8.41)	
Force-to-time ratio	6280.54 (1424.43)	6275.73 (1182.41)	6960.29 (2077.56)	6751.42 (2123.75)		6270.23 (4535.45)	6685.47 (4960.38)	6998.70 (5836.58)	6456.62 (4312.21)	

Contralateral force 1 and 2, ground reaction force exerted on the contralateral foot as the first and second peak force of the lead lag occurs, respectively.

^a $P < 0.05$ vs. 0%.

^b $P < 0.05$ vs. 10%.

^c $P < 0.05$ vs. 15%.

^d $P < 0.05$ vs. athletic bag.

of gravity of the locomotor system, thus diminishing the stability of equilibrium. The subjects were forced to adjust their gait to compensate for this change by lengthening the stance duration (or reducing the swing duration), to maintain walking stability. In the present study, although the load did not change the gait cycle duration or speed in both bags, it did change the gait pattern compared with the no load condition. The critical load that required a more stable gait pattern was smaller for the athletic bag (10% of body weight) than for the backpack (15% of body weight), which suggests that the temporal pattern of gait is more susceptible to athletic bag than backpack with increasing load for children in stair ascent.

When carrying the backpack, loads of 15% of body weight or above significantly increased the maximum (second) peak force in ascent (Table 2). This load for the athletic bag was 10% of body weight, which indicated that the peak force is sensitive to carrying methods. In the present study, children were asked to carry various loads. Mcfadyen and Winter [2] found that the second peak force was related to the push up phase, in which the ankle generated the greatest amount of energy. Carrying a load using athletic bag, with the straps crossing the right shoulder and chest with the load being positioned laterally to the left hip, causes the weight of the locomotor system to shift to the left. The sideward shifting of the load causes an asymmetrical external moment, which pulls the trunk downward and sideward. Thus, the normal trunk position is compromised to counteract the asymmetrical external load disturbance, which makes the walking unstable [10]. The unstable gait causes the extra dynamic interaction between the foot and the ground as the foot pushes up the locomotor system.

We found that the first peak force occurred immediately after the contralateral foot takeoff, at the beginning of swing (single support) phase. Our findings support those of Mcfadyen and Winter [2], who found that when the first peak force occurred, the contralateral leg started to swing and the total weight of the locomotor system was given to the lead leg while the all three joints were flexed, which caused the greatest instability. They found that the first peak force was mostly due to the extensor activities about the knee at the pull up phase, the quadriceps was quite dominant, and that the vastus lateralis was most active during this time. Zachazewski et al. [5] noted a similar phenomenon; they reported that the large power production in the hip and knee occurred at the pull up phase, while the power production in the ankle appeared at the push up phase. Although the peak force at the pull up phase is lower than that at the push up phase, the instable ipsilateral leg support causes a higher biomechanical strain in the subject's knee joint.

The higher biomechanical strain in the knee joint during stair ascent has been considered previously. Costigan et al. [6] quantified the internal load at the knee joint during stair ascent and reported that for most of the stance phase the external moment causes the knee to flex. To maintain rotational equilibrium, the knee extensors activate to counter

the rotation. The peak distal–proximal contact force was on average three times body weight and could be as high as six times body weight. The knee posterior–anterior shear forces were also higher during stair ascent than level walking. The most striking difference between stair ascent and level walking was, however, that the peak patello–femoral contact force was eight times higher during normal stair ascent. Riener et al. [3] reported that the impulsive loads experienced in normal stair ascent were over 180% of that observed for level gait. The high dynamic forces may make the natural shock absorbers of the human musculoskeletal system become insufficient. The results obtained from these two studies were based on normal stair ascent. The present study showed that carrying a load of 15% of body weight using a backpack or carrying 10% of body weight using an athletic bag for stair ascent resulted in a significantly higher peak force than for normal ascent (without load). Compared to the previous results described above, under this load condition, the significantly higher impact on the neuromuscular system, especially the knee joint, is evident.

4.2. Descent

A load of 20% of body weight induced a significant increase in gait cycle duration compared with no load (Table 1). Previous research has shown that when subjects carried different loads and walked with self-selected speed on the ground, an increase in load was associated with a decrease in the walking speed and stride length while the cadence increased with respect to an unloaded condition [11–12]. Typically, when free speed is permitted, subjects tend to minimize the effect of load on their energy cost by decreasing walking velocity. However, due to the physical constraint imposed by the fixed dimensions of the stairs, lengthening the gait cycle duration directly contributes to the reduction of propagation velocity. The data suggests that a load of 20% of body weight caused higher biomechanical stain, thus forcing the subjects to change their velocity.

As seen in Table 2, stair descent showed significantly greater first peak force than the second for both bags. As the carrying load increased from 0% to 15% of body weight, the first peak force increased from 155.74% to 195.09% of body weight. This partly confirms the findings of Adrian et al. [1], who reported 1.5 times body weight for the peak vertical force during normal stair descent, almost twice body weight for the maximum peak force during fast descent, and approximately, body weight during slow descent. The results indicated that the maximum peak force changes with walking speed and load condition.

Loy and Voloshin [4] showed that normal stair descent induced shock waves with an amplitude of 1.3 times than that observed in stair ascent and 2.5 times that experienced in level gait. Riener et al. [3] reported considerably higher ground reaction forces during descent than during ascent in the normal condition, i.e., walking without carrying a

load. When the load increased, the total weight of the locomotor system increased, which leads to an increase in shock waves. The present study agrees with previous claims by adding quantitative information obtained from the load carrying condition. Table 2 shows that in stair descent, a load of 15% of body weight was critical and significantly increased the first peak force with both bags. The maximum peak force induced by this load was 1.25 times that measured in the no load condition. Although the critical load in both ascent and descent was the same, the maximum peak force induced by this load in descent was 1.89 times that in ascent. Moreover, it was 2.19 times that with no load in ascent. Lieber [13] indicated that to sustain the shock wave that is encountered in stair descent, the antigravity muscles perform eccentric contractions, i.e. they generate force while they are lengthened. These situations are related to the potential risk of muscle fiber damage when the speed or load becomes too great. The higher impulsive cyclic loads that are generated by stair descent with a load of 15% of body weight or above may be deleterious to the musculoskeletal system in children. The continuous and daily carrying of heavy loads has been cited as an important risk factor associated with the frequency, severity, and resultant disability of low back pain and joint degeneration [14–15]. To avoid the possible hazards that are caused by stair descent, caution should be exercised in relation to the load carried, the frequency of the movement performed, and the use of shoes with proper cushioning ability.

The present study used the force-to-time ratio of the first peak force to assess the impulsive nature of the ground reaction force. Table 2 shows that there was no significant difference in the force-to-time ratio over loads in each stair mode. However, a significant difference was found between the two modes. Averaged across loads, the force-to-time ratio for descent was about three times that for ascent, which suggests that there is a much higher impulsive reaction in descent than in ascent. Previous research showed that the first peak force occurred at the loading response phase in descent, and it was dominated by the absorption of energy at both the ankle and knee immediately after the leading leg landed [5]. The high value of the force-to-time ratio in the present study further suggests that the physical action of walking down stairs results in higher dynamic loads on the children's musculoskeletal system than walking up stairs.

The reliability of the Novel Pedar System in measuring peak and mean pressure has been proved good and excellent [16–17]. However, the force values from the Novel Pedar System were consistently less than those from a force plate [16]. To enhance the accuracy, in the present study, the Novel Pedar System was calibrated every day before and after testing. It should be noted that the limitation of this analysis is that only vertical reaction forces are measured and full kinematics analysis of the lower limbs is not given.

5. Conclusion

In stair ascent, the second peak force was greater than the first peak force. The latter, however, occurred immediately after the contralateral takeoff and generated higher biomechanical demand in the unstable stance. A load of 15% of body weight induced a significant increase in maximum peak force, stance, and double support duration for backpack condition, while this load for the athletic bag was 10% of body weight. In stair descent, the first peak force was 1.59 times the second peak force. A load of 15% of body weight induced a significant increase in the maximum peak force (195% body weight) for both bags, which was 1.25 times that in the same stair mode with no load, and 1.89 and 2.19 times that in stair ascent with the same load and without a load, respectively. The force-to-time ratio in stair descent was about three times that in stair ascent.

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